

Decimal Number System → {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
Base → 10

$$\begin{aligned} (345)_{10} &\rightarrow 3 \times 10^2 + 4 \times 10^1 + 5 \times 10^0 \\ &\rightarrow 3 \times 10^2 + 4 \times 10^1 + 5 \times 10^0 \end{aligned}$$

$$(15276)_{10} \rightarrow 1 \times 10^4 + 5 \times 10^3 + 2 \times 10^2 + 7 \times 10^1 + 6 \times 10^0$$

Hexadecimal

Base → 16

{0, 1, — 9, A, B, C, D, E, F}

$$(9523)_{16} = 9 \times 16^3 + 5 \times 16^2 + 2 \times 16^1 + 3 \times 16^0$$

→ ()₁₀

$$\begin{array}{cccccc}
 4 & 3 & 2 & 1 & 0 & \\
 (A & 0 & F & 9 & 6)_{16} & \xrightarrow{\text{red}}
 \end{array}$$

$$\begin{array}{c}
 A \times 16^4 + 0 \times 16^3 + F \times 16^2 + 9 \times 16^1 + 6 \times 16^0 \\
 \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\
 10 \quad \quad \quad 15
 \end{array}$$

$$\xrightarrow{\quad \quad \quad} \quad \quad \quad)_{10}$$

④ Binary Number System $\rightarrow \{0, 1\}$ base $\rightarrow 2$

$(01(2)1)_2$ $\rightarrow$ INVALID!

④ BIN $\rightarrow$ DEC

$$\begin{array}{cccc}
 3 & 2 & 1 & 0 \\
 (1 & 1 & 0 & 1)_2
 \end{array}
 \rightarrow 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= \frac{8 + 4 + 0 + 1}{\downarrow} (13)_{10}$$

$$\begin{array}{cccccc}
 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 (1 & 0 & 1 & 0 & 1 & 0 & 1)
 \end{array}$$

$$2^6 + 2^4 + 2^2 + 2^0$$

$$64 + 16 + 4 + 1$$

$$= 85$$

$$\begin{array}{cccccc}
 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 | & 0 & 1 & 1 & 0 & 1 & 0
 \end{array}$$

$$\begin{array}{c}
 \downarrow \\
 2^6 + 2^4 + 2^3 + 2^1 \\
 \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 64 + 16 + 8 + 2 \\
 \hline
 90
 \end{array}$$

④ DEC $\rightarrow$ BIN

$(25)_{10} \rightarrow (\quad)_2$

sum

2	25	1
2	12	0
2	6	0
2	3	1
2	1	1
	0	

$(11001)_2$

$(45)_{10} \rightarrow (\quad)_2$

2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
2	1	1
	0	

$(101101)_2$

ADDITION

$$\begin{array}{r}
 2 1 0 \\
 3 6 8 \\
 + 4 5 6 \\
 \hline

 \end{array}$$

$$\begin{array}{r}
 8 \\
 + 6 \\
 \hline
 14 \\
 S = 14
 \end{array}$$

digit $\rightarrow$ $d = S \% 10 \rightarrow 4$

carry $\rightarrow$ $c = S / 10 \rightarrow 1$

$$\begin{array}{r}
 0 1 1 \\
 3 6 8 \\
 + 4 5 6 \\
 \hline
 8 2 4
 \end{array}$$

$$\begin{array}{l}
 S = 8 + 6 = 14 \\
 d = 14 \% 10 \rightarrow 4 \\
 c = 14 / 10 \rightarrow 1 \\
 \\
 S = 1 + 6 + 5 = 12 \\
 d = 12 \% 10 \rightarrow 2 \\
 c = 12 / 10 \rightarrow 1 \\
 \\
 S = 1 + 3 + 4 = 8 \\
 d = 8 \% 10 \rightarrow 8 \\
 c = 8 / 10 \rightarrow 0
 \end{array}$$

BINARY ADDITION

$$\begin{array}{r}
 01111 \\
 + 10111 \\
 \hline
 100010
 \end{array}$$

$23 \rightarrow 11 \rightarrow 34$

$$\begin{array}{l}
 d \rightarrow s/2 \\
 c \rightarrow s/2
 \end{array}$$

$$\begin{array}{l}
 1 \cdot 2 \rightarrow 1 \\
 1/2 \rightarrow 0
 \end{array}$$

$$\begin{array}{l}
 0+0 \rightarrow 00 \\
 0+1 \rightarrow 01 \\
 1+0 \rightarrow 01 \\
 1+1 \rightarrow 10
 \end{array}$$

1	1			
1	0	1	1	0
0	0	1	1	1
1	1	1	0	1

BITWISE OPERATORS

(operates on bits).

$\&$ AND

$|$ OR

$\wedge$ XOR

$\sim$ NEG

$\ll$ LEFT SHIFT

$\gg$ RIGHT SHIFT

if both bits are same
 $\rightarrow 0$
 else $\rightarrow 1$

A	B	$A \& B$	$A B$	$A \wedge B$	$\sim A$
0	0	0	0	0	1
0	1	0	1	0	1
1	0	0	1	0	0
1	1	1	1	0	0

⑨ Bitwise Operators on Numbers

5 & 6

↓
4

$$\begin{array}{r}
 \begin{array}{c} 2 \quad 1 \quad 0 \\ 5 \longrightarrow 1 \quad 0 \quad 1 \\ 6 \longrightarrow 1 \quad 1 \quad 0 \\ \hline \end{array} \\
 \text{&} \\
 \hline
 (1 \quad 0 \quad 0)_2 \longrightarrow 4
 \end{array}$$

int x = (5 & 6);
 print(x); → 4

20 | 45

$$\begin{array}{r}
 10100 \\
 101101 \\
 \hline
 111101 \longrightarrow 61 //
 \end{array}$$

20 ^ 45 →

$$\begin{array}{r}
 10100 \\
 101101 \\
 \hline
 111001 \longrightarrow 57 //
 \end{array}$$

$$\begin{matrix} 4 & 3 & 2 & 1 & 0 \\ (1 & 1 & 1 & 1 & 1) \end{matrix}_2 \rightarrow 2^4 + 2^3 + 2^2 + 2^1 + 2^0 \quad GP$$

$$2^5 - 1$$

$$a = 1, \quad t = 5 \\ r = 2$$

$$1 \left(\frac{2^5 - 1}{2 - 1} \right)$$

$$a \left[\frac{r^t - 1}{r - 1} \right]$$

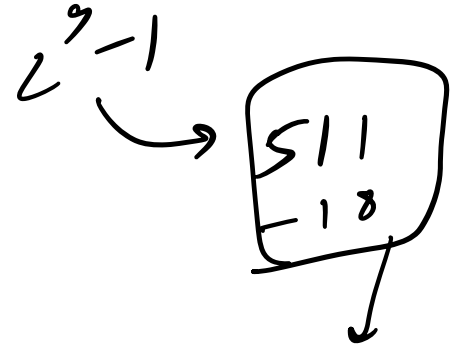
$$\begin{matrix} k & \dots & 3 & 2 & 1 & 0 \\ (1 & 1 & 1 & 1 & 1 & \dots & 1) \end{matrix}_2$$

$$2^0 + 2^1 + \dots + 2^k = 2^{k+1} - 1 \\ 1 \left[\frac{2^{k+1} - 1}{2 - 1} \right]$$

$$(1111)_2 \rightarrow 2^0 + 2^1 + 2^2 + 2^3 \\ = 2^4 - 1 \\ 16 - 1 = 15 //$$

$$\begin{matrix} 5 & 4 & 3 & 2 & 1 & 0 \\ (1 & 1 & 1 & 1 & 1 & 1) \end{matrix}_2 \rightarrow 2^0 + \dots + 2^5 \\ = 2^6 - 1 = 63 //$$

8 7 6 5 4 3 2 1 0
 1 1 1 1 0 1 1 0 1



Properties

(101010)₂
 $2^5 + 2^3 + 2^1$
 EVEN EVEN EVEN

Encryption: $2^0 \rightarrow \underline{\underline{1 : ODD}}$

Set $\rightarrow 1$
 Unset $\rightarrow 0$

5 4 3 2 1 0
 1 1 0 1 0 1

$2^5 + 2^4 + 2^2 + 2^0$
 EVEN ODD $\rightarrow$ ODD!

EVEN + EVEN
 $\downarrow$
 EVEN

EVEN + ODD
 $\downarrow$
 ODD

If a no. is ODD, it's 0th bit $\rightarrow 1$
 _____ EVEN _____ $\rightarrow 0$

$n = 5 : 101$
 $n = 15 : 1111$
 $n = 7 : 111$
 $n = 3 : 11$
 $n = 11 : 1011$

$n = 16 \rightarrow 10000$
 $n = 6 \rightarrow 110$
 $n = 4 \rightarrow 100$
 $n = 12 \rightarrow 1100$
 $n = 17 \rightarrow 11110$

1) $A \& 1$
 0

 A: EVEN
 1

 A: ODD

if $(A \% 2 == 1)$
 $\rightarrow$ ODD

else $\rightarrow$ EVEN

ODD
 A: 5

2 1 0
 1 0 1
 $\&$ 0 0 1 $\rightarrow 1$
 0 0 1 $\rightarrow 1$

A = 14

1 1 1 0
 $\&$ 0 0 0 1 $\rightarrow 1$
 0 0 0 0 $\rightarrow 0$

if $((A \& 1) == 1)$
 $\rightarrow$ ODD

else $\rightarrow$ EVEN

NOTE: PRIORITY OF BITWISE OPS
is Low!

$$\text{if } (A \& 1) == 1 \quad \checkmark$$

$$2) A \times 0 \rightarrow 0$$

$$3) A \times A \rightarrow A$$

$$4) A | 0 \rightarrow A$$

$$5) A \wedge 0 \rightarrow A$$

$$6) A \wedge A \rightarrow 0$$

$$\begin{array}{r} A : 10110 \\ \wedge A : 10110 \\ \hline A \leftarrow 00000 \end{array}$$

④ Commutative property →

$$a \times b = b \times a$$

$$a | b = b | a$$

$$a \wedge b = b \wedge a$$

$$\begin{aligned} a \times b \times c &= a \times c \times b \\ &= b \times a \times c \\ &= b \times c \times a \\ &= c \times a \times b \\ &= c \times b \times a \end{aligned}$$

XOR,
OR

④ Associative property

$$(a \times b) \times c = a \times (b \times c)$$

$$(a \mid b) \mid c = a \mid (b \mid c)$$

$$(a \wedge b) \wedge c = a \wedge (b \wedge c)$$

Q $a \wedge b \wedge a \wedge d \wedge b$

$$= \underbrace{a \wedge a}_{0} \wedge \underbrace{b \wedge b}_{0} \wedge d$$

$$\downarrow$$

$$0 \wedge 0 \wedge d$$

$$\downarrow$$

$$0 \wedge d$$

$$\downarrow$$

$$d$$

Q Given an array.

→ every element repeats twice

except one element which appears

EXACTLY ONCE!

→ find that no!

A: [2 5 3 2 5 3]

7
6 [7] 6

$$X \wedge X \rightarrow 0$$

$$= 2 \wedge 5 \wedge 3 \wedge 2 \wedge 5 \wedge 3 \wedge 6 \wedge 7 \wedge 6$$

$\begin{array}{c} 2 \wedge 2 \\ \hline \downarrow \\ 0 \end{array}$

$\begin{array}{ccccccc} & \downarrow & & & & & \\ 0 & \wedge & 0 & \wedge & 0 & \wedge & 0 \\ \hline & \downarrow & & & & & \\ & 0 & & & & & \end{array}$

$$x = 0;$$

$x = 0$
 $f(i: 0 \rightarrow N-1) \{$
 $x = x \wedge A[i];$

4

$\text{out } x_i$

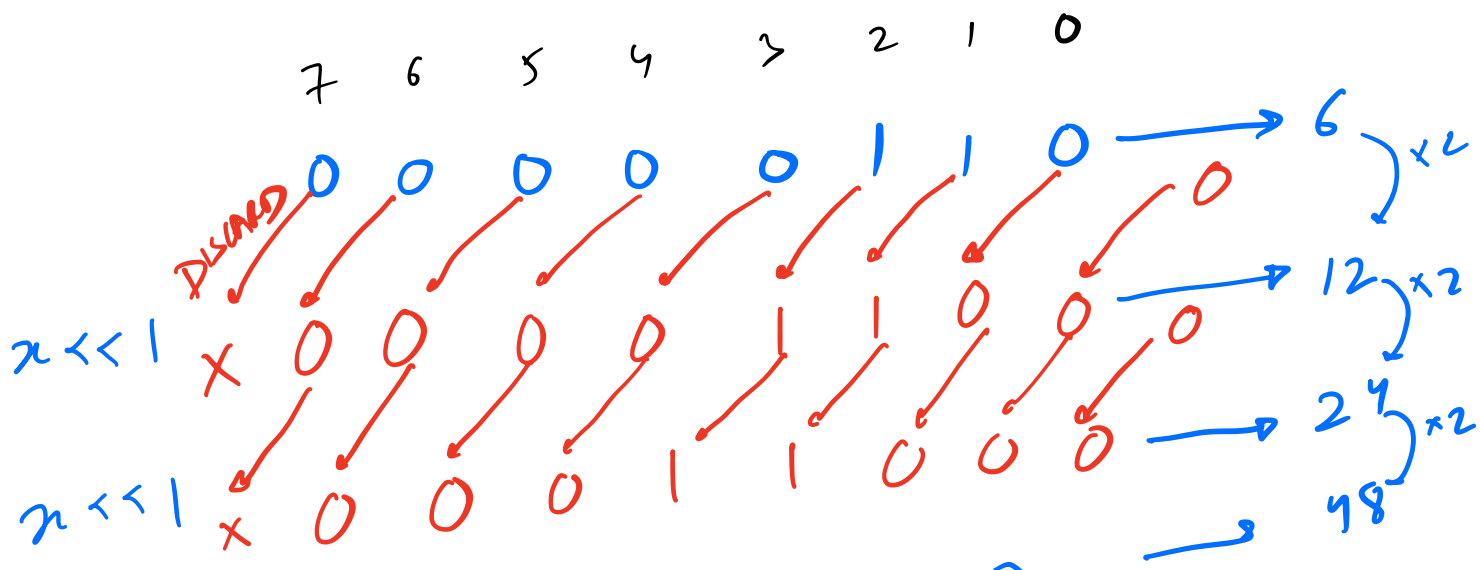
$TC = O(N)$

$SC = O(1)$

② Left Shift ($\ll$)

1 Byte = 8 bits

Byte $x = 6$;



$$\begin{aligned}
 x = 6 & : 0110 : 2^2 + 2^1 \\
 x \ll 1 & : 1100 : 2^3 + 2^2 \rightarrow 2(2^2 + 2^1)
 \end{aligned}$$

$$x \ll 1 = x \times 2$$

$$x \ll 2 = x \times 2^2$$

$$x \ll 3 = x \times 2^3$$

...

$$x \ll k = x \times 2^k$$

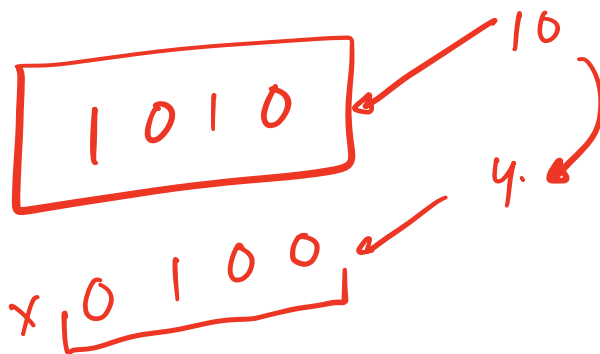
$$1 \ll k = 2^k$$

$$6 \ll 3$$

$$6 \times 2^3 \rightarrow 48$$

$$2^4 = 10000$$

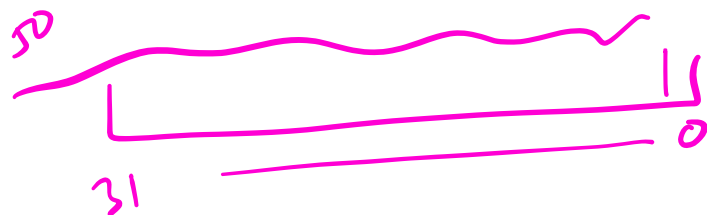
$$(2^{40}) \rightarrow (1 \ll 40) !$$



$$2^{30} \quad 1 \ll 30 \quad \checkmark$$

$$1 \ll 50 \quad \times$$

int



$$(1 \ll 50) \quad \checkmark$$

$$x = x \ll 2 \quad \checkmark$$

$$n = 6$$

$$n = n + 2$$

$$\text{print}(n) \rightarrow 6$$

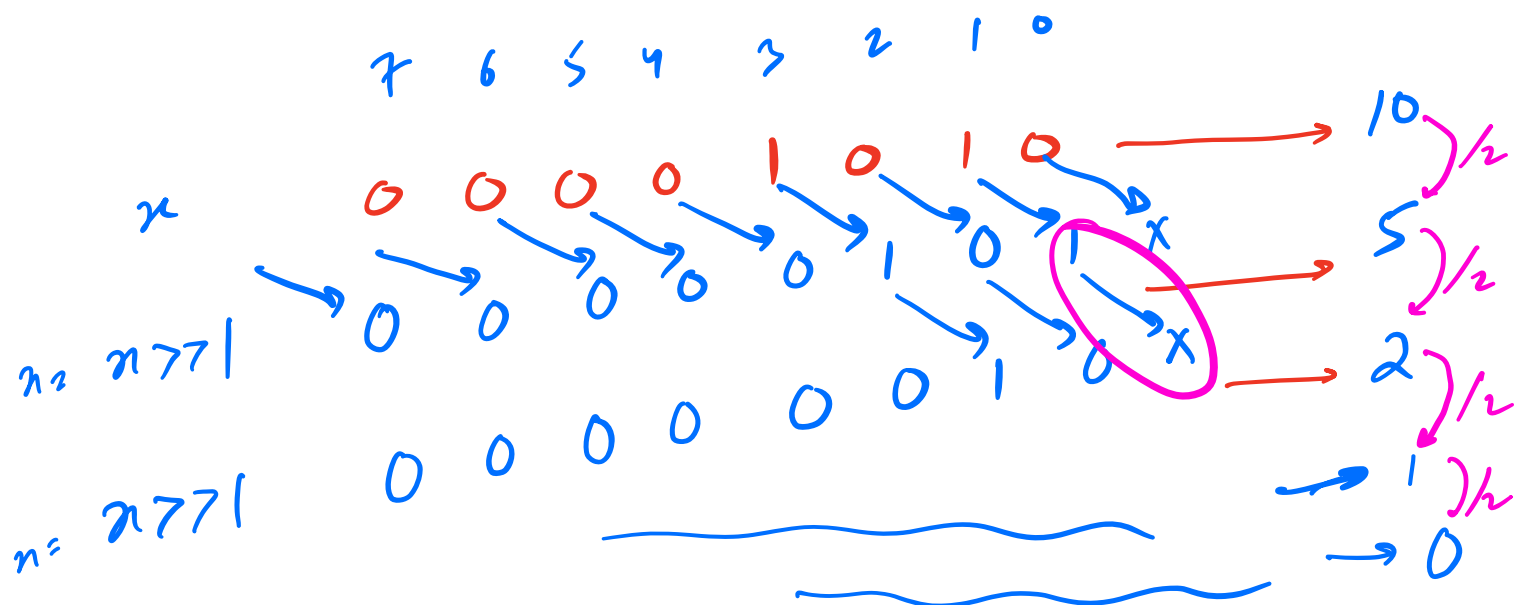
$$n = 6$$

$$n \ll 2$$

$$\text{print}(n) \rightarrow 6$$

④ Right Shift ($\gg$)

Byte $n=6$



$$10 \rightarrow 2^3 + 2^1$$

$$10 \gg 1 \rightarrow 2^2 + 2^0 \rightarrow \frac{2^3 + 2^1}{2}$$

$$n \gg 1 = \left\lfloor \frac{n}{2} \right\rfloor$$

$$n \gg 2 = \left\lfloor \frac{n}{2^2} \right\rfloor$$

$$n \gg 3 = \left\lfloor \frac{n}{2^3} \right\rfloor$$

$$n \gg k = \left\lfloor \frac{n}{2^k} \right\rfloor$$

$$25 \gg 4$$

$$\left\lfloor \frac{2^5}{2^4} \right\rfloor \left\lfloor \frac{2^5}{16} \right\rfloor$$

①

Doubt ♥

$$\begin{matrix} 3 & 2 & 1 & 0 \\ (1 & 1 & 1 & 0) \end{matrix} \quad \xrightarrow{2}$$

$$\frac{2^1 + 2^2 + 2^3}{}$$

$$a \left[\frac{x^t - 1}{x - 1} \right]$$

$$4 \left[\frac{2^3 - 1}{2 - 1} \right]$$

$$t = 3$$

$$a = 2$$

$$x = 2$$

$$2[7] \rightarrow 14$$